

DNA Spiral — English Worksheet

Topic: Lists and `%` make a spiral · **Course:** 3D Coordinates · **Level:** Extension · **Time:** about 40 minutes

TYPE `DEL` BEFORE EVERY `RUN`. THIS BUILD IS 30 TALL.

Every `???` is one short thing for you to work out. There are no answers on this sheet.

1 · Eight spots

Two lists hold eight places around a circle.

```
XS = [ 6, 4, 0, -4, -6, -4, 0, 4]
ZS = [ 0, 4, 6, 4, 0, -4, -6, -4]
```

Spot 0 is `XS[0]`, `ZS[0]`. Write it.

Write spot 2 and spot 5.

The centre is (8, 8). Spot 0 lands at $x = 8 + 6 = 14$. Where does spot 4 land?

GOT IT RIGHT IF

spot 0 and spot 4 are on opposite sides of the centre.

2 · The % sign ↺

% gives the **remainder** after dividing.

Fill this in.

l	0	1	7	8	9	15	16
l % 8	0	1					

What is the biggest number `i % 8` can ever be?

Why this matters.

Your lists only have 8 spots. `i` counts to 23. Explain what % is doing for you.

3 · The other strand ↔

```
k = i % 8  
j = (k + 4) % 8
```

Fill this in.

K	0	1	2	3	4	5	6	7
J	4	5						

Why is there a second `% 8` on the `j` line? What goes wrong without it?

GOT IT RIGHT IF

every `j` is between 0 and 7.

4 · Build it

TEMPLATE

```
XS = [6, 4, 0, -4, -6, -4, 0, 4]
ZS = [0, 4, 6, 4, 0, -4, -6, -4]

def on_chat():
    for i in range(0, 24):
        k = i % 8
        j = (??? + ???) % ???
        ax = 8 + XS[k]
        az = 8 + ZS[k]
        bx = 8 + XS[???]
        bz = 8 + ZS[???]
        blocks.fill(GOLD_BLOCK,
                    pos(ax, ???, az), pos(ax + 1, ???, az + 1), REPLACE)
        blocks.fill(EMERALD_BLOCK,
                    pos(bx, ???, bz), pos(bx + 1, ???, bz + 1), REPLACE)
    player.on_chat("dna", on_chat)
```

The `y` blanks are all the same thing.

Which variable makes it climb?

WORKS WHEN

two coloured strands wind around each other, and they are never on the same side.

IT IS A STRAIGHT TOWER?

Your `y` is a fixed number instead of a variable. It must change every turn.

BOTH STRANDS IN THE SAME PLACE?

Check the `j` line. `(k + 4)` not `(k + 8)`.

5 · Add the bars 📊

Real DNA has rungs joining the two strands.

TEMPLATE · INSIDE THE LOOP

```
if i % ??? == 0:
    blocks.fill(QUARTZ_BLOCK,
                pos(ax, i, az), pos(bx, i, bz), REPLACE)
```

24 steps. You want 6 bars. What number goes in the blank?

WORKS WHEN

you can count your bars and get the number you planned.

6 · Show Your Work 📷 🎥

Record **one video**, one take, no stopping. A phone is fine. Show these in order:

1. Type **de1** , then **dna** . Walk all the way round it.
2. Show your two lists and say what one spot means.
3. Show the **k** line and explain **%** in your own words.
4. Show the **j** line and say why it is always the opposite side.
5. Show your bars and say how you chose the number.

Try to use these words: **list, remainder, opposite, loop, climb.**

Plan what you will say before you record.

Submit ✓

Send this worksheet and your video to teacher on KakaoTalk.